(a) 1 and 3 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans : (a) Statement (1) is correct. Sand is a minor mineral as defined under section 3(e) of the mines and minerals (development and regulation) Act, 1957 (MMDR Act.)
Statement (2) is not Correct; section 15 of the MMDR Act empowers the state government to make rules for regulating the grant of mineral concessions in respect of minor minerals and for purposes connected therewith. The regulation of grant of mineral concessions for minor minerals is therefore within the legislative and administrative domain of the state governments.
Statement (3) Correct as per section 15 of the MMDR Act, 1957, the state Government have been empowered to frame rules regulating the grant of quarry leases, mining leases or other mineral concessions in respect of minor minerals and for purposes connected therewith. Both as a consequence of this and the fact that the states have been granted powers to frame rules regarding illegal mining in terms of section 23C of MMDR Act, 1957, this is a subject that is entirely under the legislative and administrative jurisdiction of the states.

25. Consider the following satatements :
1. **Coal sector was nationalized by the Government of India under Indira Gandhi.**
2. **Now, coal blocks are allocated on lottery basis.**
3. **Till recently, India imported coal to meet the shortages of domestic supply, but now India is self-sufficient in coal production.**

Which of the statements given above is/are correct?
(a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans : (a) Indira Gandhi nationalised all coal mining blocks in the early 1970s. Right to extract coal from demarcated blocks is offered on the basis of an auction, not the lottery, despite having large reserves of coal. Indian coal quality is not very good. Also, the supply lines are not very efficient and thus, many power plants in India import coal. E.g. Indonesia Supplies thermal coal to power plants run by Tata's and Adanis. Thus India is not self -reliant in coal production.

26. Recently, there was a growing awareness in our country about the importance of Himalayan nettle *(Girardinia diversifolia)* because it is found to be a sustainable source of
(a) anti-malarial drug (b) biodiesel
(c) pulp for paper industry (d) textile fibre

Ans : (d) In many parts of the world, natural resources are the only sources of livelihood opportunities available to people. Recently, Girardinia diversifolia (Himalayan nettle), a fibre yielding plant, has become an important livelihood option for people living in the remote mountainous village of the Hindu Khush Himalaya. There is a community in khar, a hamlet in Darchula district in far-western Nepal, which produces Fabrics from Himalayan nettle. The Fabric and the things made from it are sold in local as well as national and international markets as high- end products.

WORLD GEOGRAPHY

• Universe & Solar System

27. On 21st June, the Sun
(a) does not set below the horizon at the Arctic Circle
(b) does not set below the horizon at Antarctic Circle
(c) shines vertically overhead at noon on the Equator
(d) shines vertically overhead at the Tropic of Capricorn

Ans : (a) On 21st June, the sun is directly over tropic of cancer. Geometrically, it means the sun is normal (90º) to the tropic of cancer and makes an angle of 47º to the arctic circle. As such, the sun is above the horizon for all areas north of Arctic circle. The entire area in the Arctic circle experiences day. The north pole is inclined towards the sun and the places beyond the Arctic circle experience continuous daylight for about six months. Since a large portion of the Northern Hemisphere is getting light from the sun, it is summer in the regions north of the equator. The longest day and the shortest night of these places occur on 21st June.
At this time in the Southern Hemisphere all these conditions are reversed. It is winter season there. The nights are longer than days. This position of the earth is called the summer solstice.

• World Map

28. Consider the following pairs :

Sea	Bordering country
1. Adriatic Sea	Albania
2. Black Sea	Croatia
3. Caspian Sea	Kazakhstan
4. Mediterranean Sea	Morocco
5. Red Sea	Syria

Which of the pairs given above are correctly matched?
(a) 1, 2 and 4 only (b) 1, 3 and 4 only
(c) 2 and 5 only (d) 1, 2, 3, 4 and 5

Ans : (b) Croatia is not bordering the Black Sea.
The Countries bordering the Adriatic sea –Italy, Slovenia, Croatia, Bosnia, Montenegro, Albania, Greece.
Countries bordering the Black Sea- From west Bulgaria, Romania, Russia to the north-east and Ukraine and Turkey to the south.
Countries bordering the Caspian Sea- Russia, Iran, Azerbaijan, Kazakhstan and Turkmenistan.
Countries bordering the Red Sea- Saudi Arabia, Yemen, Egypt, Sudan, Eritrea, Djibouti
Countries bordering the Mediterranean Sea- Albania, Algeria, Bosnia, and Herzegnovina, Croatia, Cyprus, Egypt, France, Greece, Israel, Italy, Lebanon, Libya, Malta, Monaco, Montenegro, Morocco, Slovenia, Spain, Syria, Tunisia and Turkey.

• Atmosphere

29. In the context of which of the following do some scientists suggest the use of cirrus cloud thinning technique and the injection of sulphate aerosol into stratosphere?
(a) Creating the artificial rains in some regions
(b) Reducing the frequency and intensity of tropical cyclones
(c) Reducing the adverse effects of solar wind on the Earth
(d) Reducing the global warming

Ans : (d) Thinning cirrus cloud would be achieved by injecting ice nuclei (such as dust) into regions where cirrus clouds form, making the ice crystals bigger and reducing the cirrus optical depth thinning the clouds allow more heat to escape into space and thereby cool the planet.
Stratospheric sulfate aerosol injection has been proposed to counteract anthropogenic greenhouse gas warming and prevent regional climate emergencies. It increases the reflectivity of the lower stratosphere to cast a small proportion of the inbound sunlight back into space and cool the plan.

INDIAN CONSTITUTION & POLITY

• Citizenship, Fundamental Rights Fundamental Duties, Directive Principles of State Policies

30. In the context of polity, which one of the following would you accept as the most appropriate definition of liberty?
(a) Protection against the tyranny of political rulers
(b) Absence of restraint
(c) Opportunity to do whatever one likes
(d) Opportunity to develop oneself fully

Ans : (d) The term liberty means the absence of restraints on the activities of individuals and at the same time, providing opportunities for the development of individual personalities.
Liberty, as elaborated in the preamble is very essential for the successful functioning of the Indian democratic system. However, liberty does not mean 'license' to do what one lies and has to be enjoyed within the limitations mentioned in the constitution itself. In brief, the liberty conceived by the preamble or fundamental rights is not absolute but qualified. Hence in the above question, the most appropriate definition of liberty is to provide the opportunity to develop oneself fully.

31. Which Article of the Constitution of India safeguards one's right to marry the person of one's choice?
(a) Article 19 (b) Article 21
(c) Article 25 (d) Article 29

Ans : (b) The right to marry a person of one's choice is integral to Article 21 (right to life and liberty) of the constitution. Right to marriage is also stated under Human Rights Charter within the meaning of right to marry is a universal right and it is available to everyone irrespective of their gender. Various courts across the country have also interpreted right to marry as an integral post of right to life under Article 21. The SC has reaffirmed this in the recent ruling in the Hadiya case.

• Executive of Centre and State, Emergency Provision, Committees/ Commissions

32. In India, which of the following review the Independent regulators in sectors like telecommunications, insurance, electricity, etc?
1. **Ad Hoc Committees set up by the Parliament.**
2. **Parliamentary Department Related Standing Committees**
3. **Financial commission**
4. **Financial commission Sector Legislative Reforms Commission**
5. **NITI Aayog**

Select the correct answer using the code given below.
(a) 1 and 2 (b) 1, 3 and 4
(c) 3, 4 and 5 (d) 2 and 5

Ans : (a) The parliamentary committees are established to study and deal with various matter that cannot be directly handled by a legislature due to their volume. They also monitor the functioning of the executive branch.
The Parliamentary committees are of two kinds- standing or permanent committees and Ad-hoc committees. The former are elected or appointed periodically and they work on a continuous basis. The latter are created on an ad hoc basis as the need arises and they are dissolved after they complete the task assigned to them. They are responsible for reviewing the independent regulators in sectors like telecommunications, insurance electricity, etc.

33. Consider the following statements :
1. **The Parliament (Prevention of Disqualification) Act, 1959 exempts several posts from disqualification on the grounds of 'Office of Profit'**
2. **The above-mentioned Act was amended five times**
3. **The term 'Office of Profit' is well defined in the Constitution of India**

Which of the statements given above is/are correct
(a) 1 and 2 only (b) 3 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans : (a) Temporary suspension of disqualification in certain cases under the Parliament (Prevention of disqualification) Act, 1959 if a person being a member of parliament who immediately before the commencement of this Acts held an office of profit declared by any law repealed by this Act not to disqualify the holder thereof for being such member, becomes so disqualified by reason of any of the provision contained in this Act, such office shall not if held by such person for any period not exceeding beyond a period of six months from the commencement of this Act disqualify him for being a member of Parliament. It has been amended 5 times till date i.e. in 1993, 1999, 2000, 2006 and 2013.
The office of profit is not defined in the constitution. Rather its meaning is ascertained through Supreme Court Judgments and Election commission's guidelines.